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Counting Reinfections in SIS / SIRS Epidemics

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Motivation

In endemic diseases (SIS, SIRS), the susceptible class is heterogeneous: some individuals have never been infected, while others are recovered susceptibles who may already have transmitted the pathogen to part of their local neighbourhood. Standard EBCM lumps both into a single `S` compartment, which discards information about heterogeneous risk and can bias predictions of prevalence, time-to-equilibrium, and outbreak size in repeated waves.

@KeelingEPS2016 (PLOS Comput Biol 13(6): e1005296) propose a simple remedy: track the number of times each individual has been infected, and stratify both `S` and `I` by that count. With L strata one obtains $L + 1$ susceptible classes $S_0, S_1, \dots, S_L$ and L infectious classes $I_1, \dots, I_L$ (I_L is a saturating bucket). The dynamics are mean-field on the per-capita hazards but the partition is exactly recoverable from the underlying simulation, providing a much finer-grained picture of risk.

EdgeBasedModels.jl exposes this construction through

Symbol	Purpose
`with_reinfection_counting(prog, L)`	Lift a `DiseaseProgression` into a layered version.
`build_sis_reinfection(pgf, β, γ, L)`	Direct builder returning an EBCM SIS system with reinfection strata.
`base_compartment_of(:S_2)`□`:S`	Map a lifted name back to its base.
`infection_count_of(:I_3)`□`3`	Recover the stratum index from a lifted name.
`reinfection_totals(sol, model)`	Aggregate strata back into base compartments.

The aggregated dynamics agree with the standard `build_sis` to machine precision (the stratification is layered over the same EBCM closure).

Setup

```
using EdgeBasedModels
using OrdinaryDiffEq
using ModelingToolkit
using NetworkOutbreaks
using Graphs
using StableRNGs
using StatsPlots
```

A reinfection-counted SIS model

We build an SIS model on a 5-regular configuration-model network with transmission rate derived so the baseline has $R_0 = 2$, recovery rate $\gamma = 0.25$, and $L = 3$ reinfection strata.

```
# Universal anchors: γ=0.25, R₀=2, β derived per scenario (see plan.md)
regular_pgf(k::Int) = polynomial_pgf(vcat(zeros(k), [1.0]))
k = 5
γ = 0.25
R0_target = 2.0
T = R0_target / (k - 1)
β = T * γ / (1 - T)
pgf = regular_pgf(k)
L    = 3

system = build_sis_reinfection(pgf, β, γ, L)
```

```
ic      = default_initial_conditions(system; ε = 0.01)
prob    = ODEProblem(system.system, ic, (0.0, 40.0))
sol     = solve(prob, Tsit5(); saveat = 0.25)
ts      = sol.t
```

```
127-element Vector{Float64}:
  0.0
  0.25
  0.5
  0.75
  1.0
  1.25
  1.5
  1.75
  2.0
  2.25
  ⋮
 29.5
 29.75
 30.0
 30.25
 30.5
 30.75
 31.0
 31.25
 31.5
```

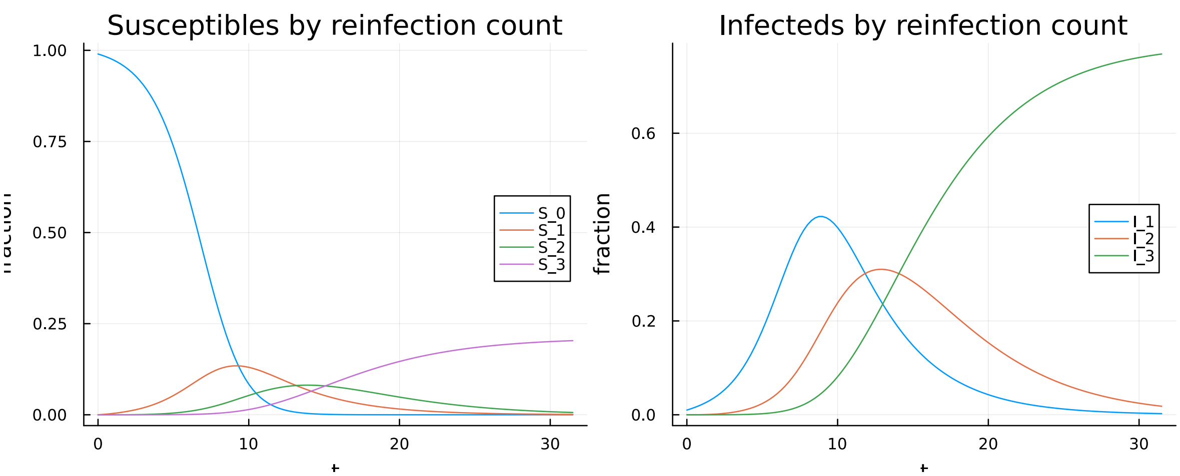
`build_sis_reinfection` constructs an `EdgeModelSystem` whose state contains the standard EBCM variables (θ, `S`, `I`) plus the per-stratum variables `S_0, …, S_L, I_1, …, I_L`.

Stratum-by-stratum susceptibles and infecteds

```
plt_S = plot(title = "Susceptibles by reinfection count", xlabel = "t",
             ylabel = "fraction", legend = :right)
for p in 0:L
    plot!(plt_S, ts, compartment(sol, system, Symbol("S_$(p)")),
          label = "S_$(p)")
end
```

```
plt_I = plot(title = "Infecteds by reinfection count", xlabel = "t",
             ylabel = "fraction", legend = :right)
for p in 1:L
    plot!(plt_I, ts, compartment(sol, system, Symbol("I_$(p)")),
          label = "I_$(p)")
end

plot(plt_S, plt_I, layout = (1, 2), size = (900, 350))
```



Two qualitative observations:

- `S_0` decays monotonically — once infected, an individual never returns to the never-infected class.
- The strata `S_p, I_p` for $p \geq 1$ rise and approach a quasi-steady state, with most mass eventually accumulating in the saturating bucket $S_L + I_L$.

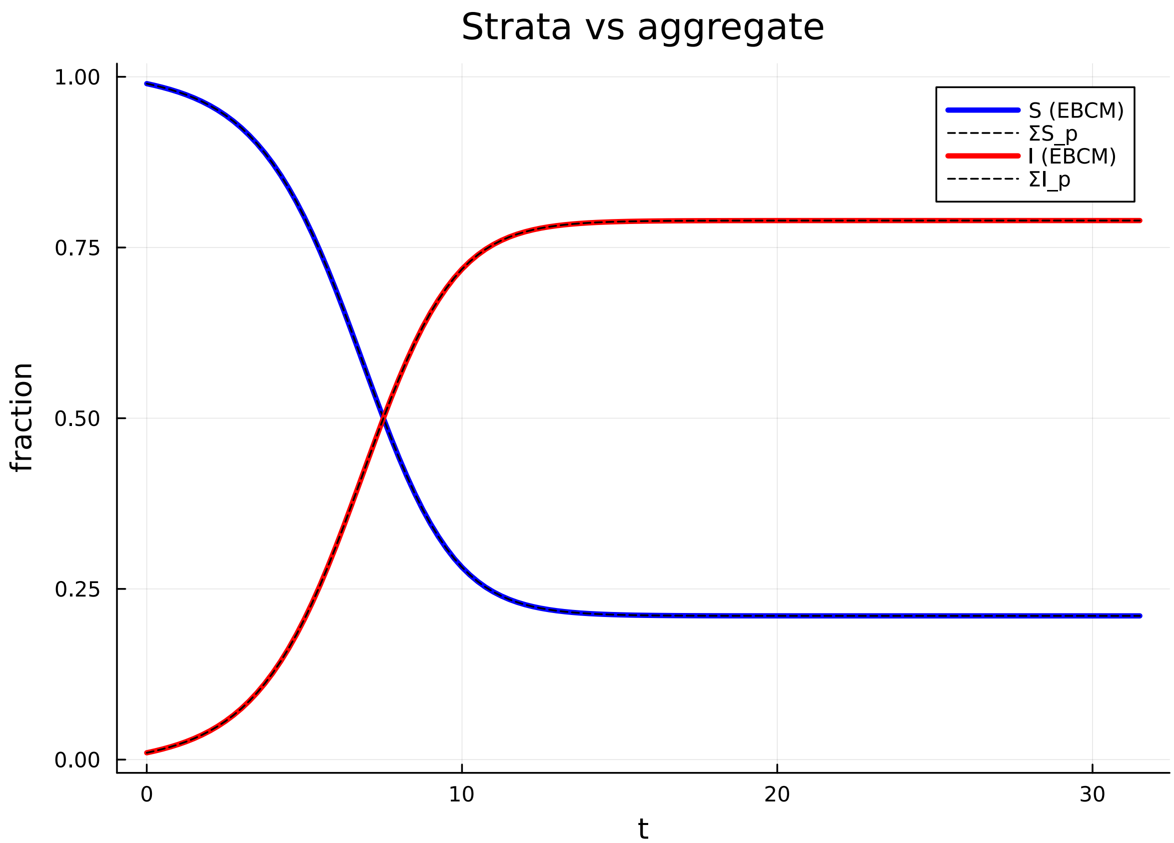
Aggregate sanity check

The strata sum to the standard EBCM aggregates:

```
S_strata = sum(compartment(sol, system, Symbol("S_$(p)")) for p in 0:L)
I_strata = sum(compartment(sol, system, Symbol("I_$(p)")) for p in 1:L)
S_agg    = compartment(sol, system, :S)
I_agg    = compartment(sol, system, :I)

plt = plot(title = "Strata vs aggregate", xlabel = "t", ylabel = "fraction")
plot!(plt, ts, S_agg,    label = "S (EBCM)", lw = 3, c = :blue)
plot!(plt, ts, S_strata, label = "ΣS_p",     ls = :dash, c = :black)
```

```
plot!(plt, ts, I_agg,    label = "I (EBCM)", lw = 3, c = :red)
plot!(plt, ts, I_strata, label = "ΣI_p",     ls = :dash, c = :black)
plt
```



The dashed (strata-summed) and solid (EBCM-aggregate) curves coincide.

Comparison against simulation

To verify that the layered closure tracks the per-stratum quantities of the simulated epidemic, we run an ensemble of SSA realisations using the companion package `NetworkOutbreaks.jl`. The simulator internally counts each per-node S⍰I event, so the histogram of infection counts is available "for free".

```
N = 1000
g = random_regular_graph(N, k; rng = StableRNG(42))

om   = OutbreakModel(sis_model(), Dict(:β ⇒ β, :γ ⇒ γ))
```

```
spec = OutbreakSpec(model = om, network = g,
                    initial = SeedFraction(:I => 0.01),
                    tspan   = (0.0, 40.0))

ens = simulate_ensemble(spec; nsims = 40, seed = 123)
```

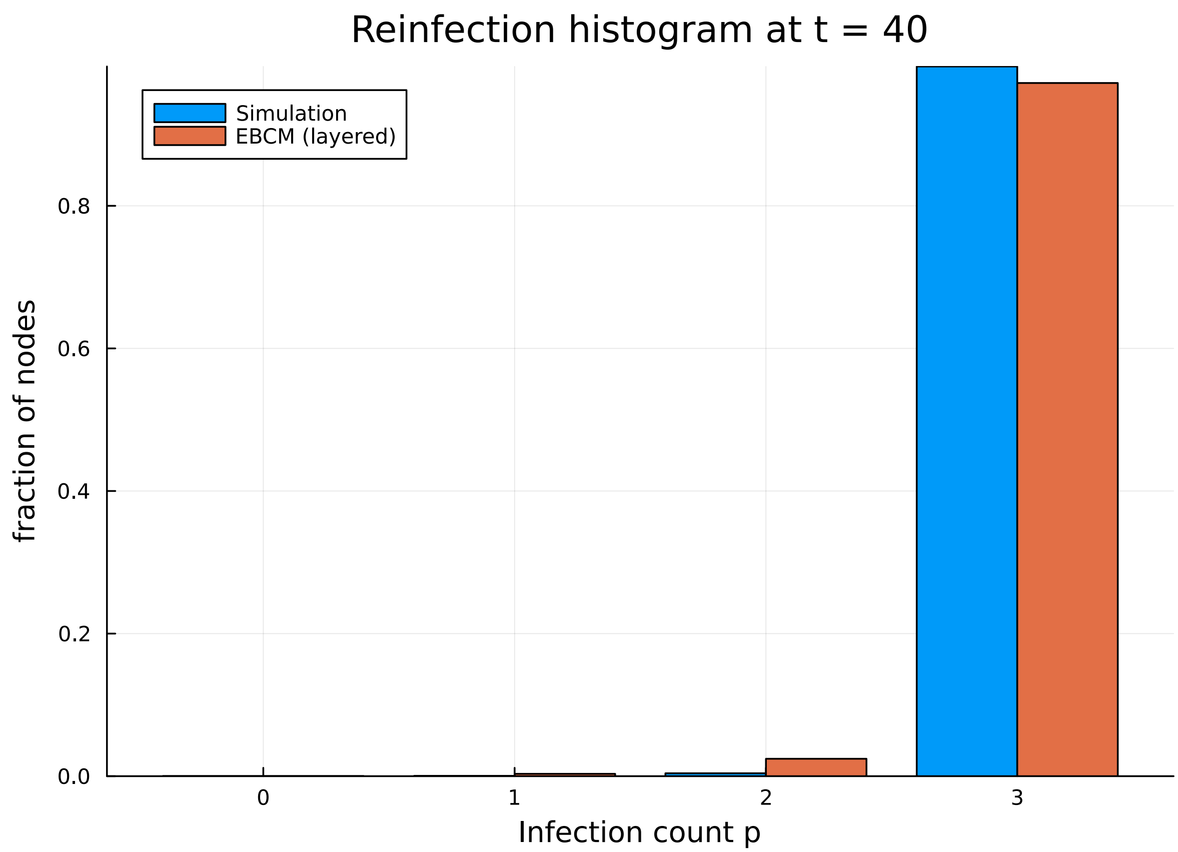
```
OutbreakEnsemble(OutbreakSpec{StaticNetwork{SimpleGraph{Int64}}}(OutbreakModel([:S, :I], Bo
```

For each trajectory we histogram the per-node infection counts at $t = 40$ and saturate counts above L:

```
counts = vcat([t.final_infection_counts for t in ens.trajectories]...)
hist   = zeros(Int, L + 1)
for c in counts
    hist[min(c, L) + 1] += 1
end
hist_frac = hist ./ length(counts)

# Predicted partition from the closure at t=30: total fraction of nodes
# whose lifetime infection count equals p (= S_p + I_p, with I_0 ≡ 0).
pred = Float64[]
for p in 0:L
    s_p = compartment(sol, system, Symbol("S_$(p)"))[end]
    i_p = p == 0 ? 0.0 : compartment(sol, system, Symbol("I_$(p)"))[end]
    push!(pred, s_p + i_p)
end

groupedbar([hist_frac pred],
    label = ["Simulation" "EBCM (layered)"],
    xlabel = "Infection count p", ylabel = "fraction of nodes",
    title = "Reinfection histogram at t = 40",
    bar_position = :dodge,
    xticks = (1:(L+1), string.(0:L)))
```



The closure is a mean-field approximation, so we expect agreement only in the bulk of the distribution; the tail in the saturating bucket is typically heavier in simulation.

When does the layered closure work well?

The per-stratum hazards are induced by the EBCM closure ($h(t) = \beta\,\phi_I\,\psi'(\theta)/\psi(\theta)$ for infection and $r(t) = \gamma\,\psi'(\theta)\,(1-\theta)/(1-\psi(\theta))$ for recovery), so the same regimes that make EBCM accurate make the reinfection counting accurate:

- configuration-model-like networks (low clustering, low degree correlation),
- large network size ($N \gtrsim 10^3$),
- many independent realisations.

For clustered or assortative networks, see the dedicated multi-type and clustering vignettes; both extensions can be combined with reinfection counting since `with_reinfection_counting` operates on the `DiseaseProgression`, not the network closure.

References

- Keeling MJ, Eames KTD, Read JM (2016). *Networks and the Epidemiology of Infectious Disease*, PLOS Comput Biol 13(6): e1005296.
- Miller JC (2011). *A note on a paper by Erik Volz: SIR dynamics in random networks.* J Math Biol 62(3): 349–358.